Note : Expression of Interest		
Rédaction : Arnaud Gruet PMO ASC	Palaiseau le :	03/03/2026

Project "The Living SysML V2 Blueprint : A Virtual Machine to Empower the Engineers of the Future"

To whom it may concern,

By this letter, I express the interest and support of the Chaire "Architecture des Systèmes Complexes" (ASC) in the project *"The Living SysML v2 Blueprint: A Virtual Machine to Empower the Engineers of the Future"*, a Franco-Brazilian proposal submitted to ANR.

The project's objective to develop a SysML V2 virtual machine with unified execution and native formal verification capabilities is fully aligned with the Chaire ASC's mission to advance rigorous, model-based approaches for the engineering of complex cyber-physical systems and systems of systems. In particular, the proposed integration of model-exploration and formal methods directly contributes to strengthening traceability, consistency, and verification across engineering environments.

From the perspective of the Chaire ASC, this initiative represents a significant step toward enhancing methodological foundations in Model-Based Systems Engineering (MBSE), while fostering closer articulation between academic research and industrial practice.
The ability to analyze behavioral properties within SysML V2 models addresses key scientific and technological challenges in the trustworthy development of complex systems.
The Franco-Brazilian international collaboration contributes to the consolidation of a robust and shared research ecosystem around next generation MBSE environments.

Subject to the usual conditions of non-interference with ongoing activities and protection of intellectual property, the Chaire ASC is willing to support the project through scientific exchanges, participation in discussions, and contribution to dissemination (members of the chair and sponsors) and academic valorization activities.

We believe that this project will contribute meaningfully to the structuring of future engineering practices and to the development of robust, verifiable model-based ecosystems.

Arnaud GRUET
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PMO of the chair « Architecture des Systèmes Complexes"